Spatial Economics - TA Session 2

Model Estimation and Counterfactuals with Caliendo et al. (2018)

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Framework Recap

Consider an economy with

- parameters $\Sigma = \{\sigma, \alpha, \{d_{ni}\}_{i \in N, n \in N}\}$
- (exogenous) fundamentals $A = \{A_i, H_i\}_{i \in N}$
- (endogenous) allocations $Y = \{L_i\}_{i \in N}$
- (endogenous) prices $P = \{w_i\}_{i \in N}$
- agents' optimality conditions and market clearing $F(\Sigma, A, Y, P) = 0$
- equilibrium defined as:

Competitive Equilibrium (CE)

Given parameters and fundamentals $\{\Sigma, A\}$, a CE is a set of prices and allocations $\{P, Y\}$ such that given prices P, allocations Y solve $F(\Sigma, A, Y, P) = 0$.

Framework Recap

- Solving the model: characterize endogenous Y, P as a function of fundamentals

$$\Sigma \quad + \quad A \quad \xrightarrow{F} \quad Y, P$$

- Calibration: find Σ that is established by the literature or matches model implied moments

$$\underbrace{Y, P}_{\text{observed}} \quad \xrightarrow{F} \quad \underbrace{\Sigma}_{\text{estimated}}$$

- Inversion: find A that rationalizes data under the assumptions of the model and Σ

$$\Sigma \quad + \quad \underbrace{Y, P}_{\text{observed}} \quad \xrightarrow{F} \quad \underbrace{A}_{\text{estimated/inverted}}$$

- Counterfactuals: ask how changes in exogenous objects would change the outcome

$$\underbrace{\hat{\Sigma} \quad + \quad \hat{A}}_{\text{assumed changes}} \quad \xrightarrow{F} \quad \underbrace{\hat{Y}, \hat{P}}_{\text{object of interest}}$$

Regional $\times$ Sectoral Trade and Migration Model (Caliendo et al., 2018)

Question: How does elasticities of TFP, GDP, employment with respect to regional/sectoral productivity changes differ at aggregate, regional, sectoral, regional$\times$sectoral level?

- Model with 50 US States n, 26 sectors j, labor L_n^j and land H_n^j as factors
- Labor is mobile across regions and sectors, land is mobile across sectors
- In each j, intermediate varieties $\omega^j \in [0,1]^j$ produced by representative firms $\{\omega_n^j\}_n$
  - For each ω^j, each firm ω_n^j draws idiosynratic productivity z_n^j
  - $\{\omega_n^j\}_n$ compete to sell variety ω^j to final good-j aggregators Q_n^j across n
  - Only ω_n^j with the lowest MC (incl. trade cost κ_{mn}) among $\{\omega_n^j\}_n$ sell ω^j to Q_m^j
- Final good-j aggregators produce Q_n^j by aggregating varieties $\omega^j \in [0,1]^j$, ensuring each ω^j is supplied from the cheapest among $\{\omega_n^j\}_n$

Intermediate Firms - Allocations (Caliendo et al., 2018)

Choose a j, and choose a $\omega^j \in [0,1]^j$. Producer of ω^j in n has:

$$q_n^j\left(z_n^j\right) = z_n^j \left[T_n^j \left[h_n^j\left(z_n^j\right)\right]^{\beta_n} \left[l_n^j\left(z_n^j\right)\right]^{(1-\beta_n)}\right]^{\gamma_n^j} \prod_{k=1}^{J} \left[M_n^{jk}\left(z_n^j\right)\right]^{\gamma_n^{jk}}$$

- z_n^j : ω_n^j's productivity draw where $\Pr(z_n^j \leq z) = F_n^j(z) = e^{-z^{-\theta^j}}$
- $q_n^j(z_n^j)$: production technology of ω_n^j with z_n^j
- T_n^j : fundamental productivity, common to all intermediate firms in n, j
- $h_n^j(z_n^j)$: input use of land by i with z_n^j
- $l_n^j(z_n^j)$: input use of labor by i with z_n^j
- $M_n^{jk}(z_n^j)$: input use of final good (locally) produced by sector k by i with z_n^j

Intermediate Firms - Parameters (Caliendo et al., 2018)

$$\underbrace{q_n^j\left(z_n^j\right)}_{\text{output}} = \underbrace{z_n^j\left[T_n^j\left[h_n^j\left(z_n^j\right)\right]^{\beta_n}\left[l_n^j\left(z_n^j\right)\right]^{(1-\beta_n)}\right]^{\gamma_n^j}}_{\text{value added (sums to GDP)}} \underbrace{\prod_{k=1}^{J}\left[M_n^{jk}\left(z_n^j\right)\right]^{\gamma_n^{jk}}}_{\text{inputs}}$$

- β_n : share of land/structures in value added (from BEA, GDP by state)
- γ_n^j : share of value added in output for n, j (from BEA, input output (io) tables)
- γ_n^{jk} : input share of sector k in output of sector j produced in n (from BEA, io tables)
- θ^j : Frechet shape, also elasticity of trade to trade cost (from Caliendo and Parro (2015))

In replication package, θ^j is $1/T$, β_n is B. T_n^j are not calculated since model is not inverted.

Intermediate Firms - Price (Caliendo et al., 2018)

- Fix a variety ω^j. Producer ω^j_n has the marginal cost given by

$$\frac{x^j_n}{z^j_n \left(T^j_n\right)^{\gamma^j_n}} \quad \text{where} \quad x^j_n = B^j_n \left[r_n^{\beta_n} w_n^{1-\beta_n}\right]^{\gamma^j_n} \prod_{k=1}^{J} \left[P^k_n\right]^{\gamma^{jk}_n}$$

and $B^j_n = \left[\gamma^j_n \left(1-\beta_n\right)^{(1-\beta_n)} \beta_n^{\beta_n}\right]^{-\gamma^j_n} \prod_{k=1}^{J} \left[\gamma^{jk}_n\right]^{-\gamma^{jk}_n}$

- Then, price of ω^j_n, when sold to country m is $\frac{\kappa^j_{mn} x^j_n}{z^j_n \left(T^j_n\right)^{\gamma^j_n}}$

- And, price of ω^j, in country m is $\min_{n \in \{1,\ldots,N\}} \frac{\kappa^j_{mn} x^j_n}{z^j_n \left(T^j_n\right)^{\gamma^j_n}}$

Final Good Firms (Caliendo et al., 2018)

- A final good firm in each n, j aggregates across varieties $u^j \in [0,1]^j$

$$Q_n^j = \left[\int_0^1 \tilde{q}_n^j\left(u^j\right)^{1-1/\eta_n^j} du^j\right]^{\eta_n^j/\left(\eta_n^j-1\right)}$$

- Change of variables to instead integrate across vectors of $z^j = [z_1^j, ..., z_N^j]$
- $\tilde{q}_n^j\left(z^j\right)^{1-1/\eta_n^j}$ is weighted by how many varieties u^j exist with that z^j

$$Q_n^j = \left[\int \cdots \int \tilde{q}_n^j\left(z^j\right)^{1-1/\eta_n^j} \phi^j\left(z^j\right) dz_1^j \cdots dz_N^j\right]^{\eta_n^j/\left(\eta_n^j-1\right)}$$

- with ideal price index $P_n^j = \left[\int p_n^j\left(z^j\right)^{1-\eta_n^j} \phi^j\left(z^j\right) dz^j\right]^{1/\left(1-\eta_n^j\right)}$

Derivation From (Eaton and Kortum, 2002)

Use price index, unit cost, and Frechet trick.

$$
\begin{aligned}
P_n^j &= \left[\int p_n^j\left(z^j\right)^{1-\eta_n^j} \phi^j\left(z^j\right) dz^j\right]^{1/\left(1-\eta_n^j\right)} \\
&= \left[\int \left(\min_i \left\{\left(\kappa_{ni}^j x_i^j\right) / \left(z_i^j (T_i^j)^{\gamma_i^j}\right)\right\}\right)^{1-\eta_n^j} \phi^j\left(z^j\right) dz^j\right]^{1/\left(1-\eta_n^j\right)} \\
&= \cdots \\
&= \Gamma\left(\xi_n^j\right)^{\frac{1}{1-\eta_n^j}} \left[\sum_{i=1}^{N} \left[x_i^j \kappa_{ni}^j\right]^{-\theta^j} \left(T_i^j\right)^{\theta^j \gamma_i^j}\right]^{-\frac{1}{\theta^j}} \quad \text{if } j \text{ tradable} \\
&\quad \Gamma\left(\xi_n^j\right)^{\frac{1}{1-\eta_n^j}} x_n^j \left(T_n^j\right)^{-\gamma_n^j} \quad \text{otherwise}
\end{aligned}
$$

where $\Gamma(\xi_n^j)$ is the Gamma function evaluated at $\xi_n^j = 1 + \left(1 - \eta_n^j\right) / \theta^j$.

Expenditure Shares (Eaton and Kortum, 2002)

$$\underbrace{X_{ni}^{j}}_{n\text{'s expenditure on } j,\, n} = \Pr\left[p_{ni}^{j}\left(z^{j}\right) \leq \min_{m \neq i}\left\{p_{nm}^{j}\left(z^{j}\right)\right\}\right] \underbrace{X_{n}^{j}}_{n\text{'s expenditure on } j}$$

$$\underbrace{\pi_{ni}^{j}}_{\text{bilateral trade shares}} \equiv \frac{X_{ni}^{j}}{X_{n}^{j}} = \left[\frac{\kappa_{ni}^{j} x_{i}^{j} \Gamma\left(\xi_{n}^{j}\right)^{\frac{1}{1-\eta_{n}^{j}}}}{\left(T_{i}^{j}\right)^{\gamma_{i}^{j}} P_{n}^{j}}\right]^{-\theta^{j}} = \frac{\left[x_{i}^{j}\kappa_{ni}^{j}\right]^{-\theta^{j}} T_{i}^{j\theta^{j}\gamma_{i}^{j}}}{\sum_{m=1}^{N}\left[x_{m}^{j}\kappa_{nm}^{j}\right]^{-\theta^{j}} T_{m}^{j\theta^{j}\gamma_{m}^{j}}}$$

In package, $X_{ni}^{j}, X_{n}^{j}, \pi_{ni}^{j}$ come from Commodity Flow Survey, X_{ni}^{j} is xbilat, π_{ni}^{j} is Din

Consumers

- Household in n has preferences $U\left(C_n\right) = \prod_{j=1}^{J}\left(c_n^j\right)^{\alpha^j}$ where $\sum_{j=1}^{J}\alpha^j = 1$
- Ideal price index is $P_n = \prod_{j=1}^{J}\left(P_n^j/\alpha^j\right)^{\alpha^j}$
- Income is location specific

$$I_n = \underbrace{w_n}_{\text{labor income}} + \underbrace{\chi}_{\text{US portfolio income}} + \underbrace{\left(1-\iota_n\right) r_n H_n / L_n}_{\text{local rent income}}$$

where $\chi = \sum_{i=1}^{N}\iota_i r_i H_i / \sum_{i=1}^{N} L_i$

L_n comes from BEA.
α^j's are calculated for aggregate US using GDP, imports, and exports in each sector.

Consumers

- Model allows trade imbalances. Let $\Upsilon_n \equiv \iota_n r_n H_n - \chi L_n$.
  - If $\Upsilon_n > 0$, n sends money out, so n gives trade surplus
  - Let Υ_n^{data} denote trade surplus implied by data on X_{ni}^j
  - Model implied ι_n are then estimated to minimize $|\Upsilon_n - \Upsilon_n^{data}|$
- By free market mobility

$$L_n = \frac{\left(\frac{\omega_n}{UP_n}\right)^{\frac{1}{\beta_n}} H_n}{\sum_{i=1}^N \left(\frac{\omega_i}{UP_i}\right)^{\frac{1}{\beta i}} H_i} L$$

where $\omega_n \equiv \left(\frac{w_n}{1-\beta_n}\right)^{1-\beta_n} \left(\frac{r_n}{\beta_n}\right)^{\beta_n}$ is the unobserved local composite factor price

I_n are calculated as $I_n = VA_n^{data} / L_n - \Upsilon_n / L_n$ where VA_n^{data}, value added, comes from BEA.

Other Equilibrium Conditions (Check the paper on how to take care of S_n)

- Regional Market Clearing

$$X_n^j = \sum_k \gamma_n^{kj} \sum_i \pi_{in}^k X_i^k + \alpha^j \left(\omega_n \left(H_n \right)^{\beta_n} \left(L_n \right)^{1-\beta_n} - \Upsilon_n - S_n \right)$$

- and Trade Balance

$$\sum_{j=1}^{J} X_n^j + \Upsilon_n + S_n = \sum_{j=1}^{J} \sum_{i=1}^{N} \pi_{in}^j X_i^j$$

- imply

$$\omega_n \left(H_n \right)^{\beta_n} \left(L_n \right)^{1-\beta_n} = \sum_j \gamma_n^j \sum_{i=1}^{N} \pi_{in}^j X_i^j$$

Hat Algebra

$$\hat{x}_n^j = (\hat{\omega}_n)^{\gamma_n^j} \prod_{k=1}^{J} \left(\hat{P}_n^k\right)^{\gamma_n^{jk}} \tag{1}$$

$$\hat{P}_n^j = \left(\sum_{i=1}^{N} \pi_{ni}^j \left[\hat{\kappa}_{ni}^j \hat{x}_i^j\right]^{-\theta^j} \hat{T}_i^{j\theta^j} \gamma_i^j \right)^{-1/\theta^j} \tag{2}$$

$$\pi_{ni}^{j'} = \pi_{ni}^j \left(\frac{\hat{x}_i^j}{\hat{P}_n^j} \hat{\kappa}_{ni}^j \right)^{-\theta^j} \hat{T}_i^{j\theta^j \gamma_i^j} \tag{3}$$

$$\hat{L}_n = \frac{\left(\frac{\hat{\omega}_n}{\varphi_n \hat{P}_n \hat{U} + (1-\varphi_n)\hat{b}_n}\right)^{1/\beta_n}}{\sum_i L_i \left(\frac{\hat{\omega}_i}{\varphi_i \hat{P}_i \hat{U} + (1-\varphi_i)\hat{b}_i}\right)^{1/\beta_i}} L \tag{4}$$

$$X_n^{j'} = \sum_{k=1}^{J} \gamma_n^{k,j} \left(\sum_{i=1}^{N} \pi_{in}^{k\prime} X_i^{k\prime} \right) + \alpha^j \left(\hat{\omega}_n \left(\hat{L}_n\right)^{1-\beta_n} \left(I_n L_n + \Upsilon_n + S_n\right) - S_n' - \Upsilon_n' \right) \tag{5}$$

$$\hat{\omega}_n \left(\hat{L}_n\right)^{1-\beta_n} \left(L_n I_n + \Upsilon_n + S_n\right) = \sum_j \gamma_n^j \sum_i \pi_{in}^{\prime j} X_i^{\prime j} \tag{6}$$

Roadmap - "Solving" the baseline equilibrium

- Solving the model in levels requires knowing fundamentals: T_i^j, κ_{ni}^j
- Authors don't invert the model
- Express the equilibrium in changes and there should be no changes in steady state

$$\underbrace{\theta^j, \alpha^j, \beta_n, \gamma_n^j, \gamma_n^{jk}}_{\text{calibration}} \quad + \quad \underbrace{I_n, L_n, \Upsilon_n, \pi_{ni}^j}_{\text{data}} \quad \overset{\hat{F}}{\to} \quad \underbrace{\tilde{I}_n, \tilde{L}_n, \tilde{\Upsilon}_n, \tilde{\pi}_{ni}^j}_{\text{baseline allocations}} \quad \overset{\hat{F}}{\to} \quad \underbrace{\hat{I}_n, \hat{L}_n, \hat{\Upsilon}_n, \hat{\pi}_{ni}^j}_{\text{counterfactual equilibrium}}$$

- *CPRHS_Benchmark.m* solves model-consistent baseline allocations that fix residuals associated with trade imbalances and ι_n's
- *Shock_California_Computers.m* (many alternatives) solves for counterfactual equilibrium

Why We Need Baseline Allocations?

- $\iota_n = \arg\max |\mathrm{Y}_n - \mathrm{Y}_n^{data}|$ subject to $\iota \in [0, 1]$ does not fit the data, and lead to a residual surplus $S_n \neq 0$
- Baseline equilibrium solves for changes needed to achieve $S_n' = 0$
- Changes are small since S_n is close to 0
- This produces a new equilibrium allocation that is computed using data, but adjusted so that it is model consistent
- Counterfactuals are then computed using these allocations

Sketch of the Algorithm for Solving the Equilibrium Allocations

- Assume changes in fundamentals $\hat{\kappa}^j_{ni}$, $\hat{H}_n$, $\hat{T}^j_{ni}$
- Guess changes in the composite factor price $\hat{\omega}^{(0)}_n$
- While $\hat{\omega}^{(l)}_n - \hat{\omega}^{(l-1)}_n > \text{tol}$
  1. Compute $\hat{P}^{j(l)}_n, \hat{x}^{j(l)}_n$ using (1) and (2)
  2. Compute $\pi^{j'}_{ni}(\hat{\omega})$ using (3)
  3. Compute $\hat{L}_n(\hat{\omega})$ using (4)
  4. Compute $X^{j'}_n(\hat{\omega})$ using (5)
  5. Compute $\hat{\omega}^{(l+1)}_n$ using (6) and update $l = l + 1$
- Note that authors update $\hat{\omega}^{(l)}_n$ partially, i.e. $\hat{\omega}^{(l+1)}_n = \delta\hat{\omega}^{(l)}_n + (1-\delta)\hat{\omega}^{(new)}_n$ for $\delta \in (0,1)$

Caliendo et al. (2018) Replication Package

Download the replication package which includes the code used in this TA session at:

https://sites.google.com/site/lorenzocaliendo/research/CPRHS

References I

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